LEVEL III

BUILT-IN FUNCTIONS:

ASCII VALUES:

1. 0 to 1 -> 48 to 57
2. A to Z -> 65 to 90
3. a to z -> 97 to 122
4. Space -> 32

ARRAY: - import java.util.Arrays;

1. **int arrayName[]=new int[100];** - Array declaration

2. **arrayName.length** Length of the Array
   int len = arr.length();

3. **Arrays.sort(arrayName);** - sort in ascending order

4. **Arrays.binarySearch(arrayName, numberToSearch);** - search the number and returns it's index position
   int[] arr = {1, 3, 5, 8};
   int index = Arrays.binarySearch(arr, 5);
   // Output: index = 2

5. **Arrays.copyOf(arrayName, number_Of_Index_positions);**
   int[] arr = {1, 2, 3};
   int[] newArr = Arrays.copyOf(arr, 5);
   // Output: newArr = [1, 2, 3, 0, 0]

6. **Arrays.equals(arr1, arr2);**
   int[] arr1 = {1, 2, 3};
   int[] arr2 = {1, 2, 3};
   boolean isEqual = Arrays.equals(arr1, arr2);
   // Output: isEqual = true

7. **Arrays.fill(arrayName, Number_to_fill);**
```
int[] arr = new int[5];
Arrays.fill(arr, 7);
// Output: arr = [7, 7, 7, 7, 7]
```

8. **Arrays.copyOfRange(arrayName, first_index, second_index);**
```
int[] arr = {1, 2, 3, 4, 5};
int[] newArr = Arrays.copyOfRange(arr, 1, 4);
// Output: newArr = [2, 3, 4]
```

9. **Arrays.mismatch(arr1, arr2);**
```
int[] arr1 = {1, 2, 3, 4};
int[] arr2 = {1, 2, 0, 4};
int mismatchIndex = Arrays.mismatch(arr1, arr2);
System.out.println(mismatchIndex);
// Outputs: 2 (the first mismatch is at index 2)
```

10. **Arrays.toString(arrayName)** - Converts array to a readable string
```
int[] arr = {1, 2, 3};
String arrStr = Arrays.toString(arr);
System.out.println(arrStr);
// Output: [1, 2, 3]
```

String: - import java.lang.String;

11. **split()** - Split the words in a sentence
```
String sentence = "Java is fun";
String[] words = sentence.split(" ");
for(int i=0;i<words.length;i++){
System.out.print(words[i]+' ');
}
//Output: Java is fun
```

12. **charAt(index)** - char ch = str.charAt(0); // Returns 'H' in Hello

13. **substring(beginIndex, endIndex)**
```
String sub = str.substring(0, 5);   // Returns "Hello"  in HelloWorld
```

14. **StringName.substring(beginIndex)**

```
String sub = str.substring(7);   // Returns "rld!"  in HelloWorld!
```

15. 1stString.equals(2ndString)

```
String s1 = "Java";
String s2 = "Java";
boolean isEqual = s1.equals(s2);   // Returns true
```

16. 1stString.equalsIgnoreCase(2ndString)

```
String s1 = "Java";
String s2 = "java";
boolean isEqual = s1.equalsIgnoreCase(s2);   // Returns true
```

17. str1.concat(str2); - Concatenate two strings

```
String s1 = "Java";
String s2 = " Programming";
String result = s1.concat(s2);
System.out.print(result);
// Output: Java Programming
```

18. stringName.trim();

```
String text = "     Java is fun     ";
String trimmed = text.trim();
System.out.print(trimmed);
//Output: Java is fun (without whitespaces at the beginning & the end)
```

19. **boolean** contains = str.**contains**("ell");
// true or false (to check if a string contains a substring)

20. indexOf('char') - to find the index of a character or substring

```
String str = 'Hello';
int index = str.indexOf('l'); // 2 (first index of a specific character)
int lastIndex = str.lastIndexOf('l'); // 3   (last index of a specific character)
```

21. toUpperCase() & toLowerCase()

```
String upper = str.toUpperCase(); // "HELLO"
String lower = str.toLowerCase(); // "hello"
```

22. reverse()

```
StringBuilder rev = new StringBuilder(noSpace);
 rev.reverse();
```

23. Remove space

```
StringBuilder noSpace = new StringBuilder();
        for (int i = 0; i < str.length(); i++) {
```

```
            if (str.charAt(i) != ' ') {
                noSpace.append(str.charAt(i));
            }
        }
```

INPUT AND PRINT ARRAY

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
            for (int i = 0; i < n; i++) {
            arr[i] = obj.nextInt();
        }
        System.out.print("Output: ");
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

ALTERNATE METHOD: - (size must be declared, if undeclared additional space will be replaced with zero in the output)

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int arr[] = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = obj.nextInt();
        }
        System.out.println("Another Method");
        for (int i : arr) {
            System.out.println(i);
        }
```

```
        obj.close();
    }
}
```

1.MIN MAX

```
public class Main{
    public static void main(String[] args) {
        // Declare and initialize an array of integers
        int[] scores = {90, 70, 80, 95, 85};

        // Print the array elements
        System.out.println("Array elements:");
        for (int i = 0; i < scores.length; i++) {
            System.out.print(scores[i] + " ");
        }
        System.out.println();

        // Find the maximum score in the array
        int maxScore = scores[0];
        for (int i = 1; i < scores.length; i++) {
            if (scores[i] > maxScore) {
                maxScore = scores[i];
            }
        }
        System.out.println("Maximum score: " + maxScore);

        // Find the minimum score in the array
        int minScore = scores[0];
        for (int i = 1; i < scores.length; i++) {
            if (scores[i] < minScore) {
                minScore = scores[i];
            }
        }
        System.out.println("Minimum score: " + minScore);

    }
}
```

2.The output array should hold the same first value and

remaining value should be difference of iterative values
Input : 5 (size)
2 6 8 4 9
Output : 2 4 2 -4 5

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int arr[] = new int[100];
        int arr1[] = new int[100];

        int n = obj.nextInt();
        for(int i=0; i<n; i++){
            arr[i]=obj.nextInt();
        }
        arr1[0]=arr[0];
        for(int i=1;i<n;i++){
            arr1[i]=arr[i]-arr[i-1];
        }
        System.out.println("Array: ");
        for(int i=0;i<n;i++){
            System.out.print(arr1[i]+" ");
        }
    }
}
```

3. WORD COUNT

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String arr[]= str.split("\\s+");
        int c= arr.length;
        System.out.println(c);
    }
}
```

4. Replace by letter

```
import java.util.Scanner;
```

```
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String s2 = str.replace('a','b');
        System.out.println(s2);
    }
}
```

5. Replace by word

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String s2 = str.replace("Hello","bye");
        System.out.println(s2);
    }
}
```

6. Matrix addition

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int arr1[][] = new int[100][100];
        int arr2[][] = new int[100][100];
        int res[][] = new int[100][100];
        int r = obj.nextInt();
        int c = obj.nextInt();

        for(int i =0; i<r;i++){
            for(int j=0;j<c;j++){
                arr1[i][j]=obj.nextInt();
            }
        }
        for(int i =0; i<r;i++){
            for(int j=0;j<c;j++){
                arr2[i][j]=obj.nextInt();
            }
        }
```

```
        For(int i =0; i<r;i++){
            for(int j=0;j<c;j++){
                res[i][j]=arr1[i][j] + arr2[i][j];
            }
        }
        for(int i =0; i<r;i++){
            for(int j=0;j<c;j++){
                System.out.print(res[i][j]+" ");
            }
            System.out.println();
        }
    }
}
```

7. To split each letter in word
hello
[h, e, l, l, o]

```
import java.util.Arrays;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        char arr[] = str.toCharArray();

        // Print the array as a string
        System.out.println(Arrays.toString(arr));
    }
}
```

8. TO CHECK IF TWO ARRAYS ARE SIMILAR

```
import java.util.Arrays;
class Main
{
public static void main(String[] args)
{
Scanner obj=new Scanner(System.in);
int n;
n=obj.nextInt();
```

```
int arr[]=new int[n];
for(int i=0;i<n;i++)
{
arr[i]=obj.nextInt();
}
int n1=obj.nextInt();
int arr1[]=new int[n1];
for(int i=0;i<n1;i++)
{
arr1[i]=obj.nextInt();
}
if(Arrays.equals(arr,arr1))
{
System.out.println("TRUE");
}
else
{
System.out.println("NOT TRUE");
}
}
}
```

9. SORTING ARRAY (using built-in function)

```
import java.util.Scanner;
import java.util.Arrays;
public class Main {
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int arr[] = new int[n];
        for(int i=0;i<n;i++){
            arr[i]=obj.nextInt();

        }
        Arrays.sort(arr);
```

```
        for(int i=0;i<n;i++){
            System.out.print(arr[i]+' ');
        }
    }
}
```

10. RANK THE INPUT IN ASCENDING ORDER

```
import java.util.Scanner;
import java.lang.String;
public class Main{
    public static void main(String[] args){
        Scanner obj = new Scanner(System.in);
        int s = obj.nextInt();
        int arr[] = new int[s];
        int rank[] = new int[s];
        for(int i=0;i<s;i++){
            arr[i]=obj.nextInt();
        }
        for(int i=0;i<s;i++){
            rank[i]=s;
            for(int j=0;j<s;j++){
                if(arr[i]<arr[j]){
                    rank[i]--;
                }
            }
        }
        for(int i=0;i<s;i++){
            System.out.print(rank[i]);
            if(i<s-1){
                System.out.print(',');
            }
        }
        }
    }
```

11. PRINT THE INITIAL LETTERS IN UPPER-CASE
INPUT: “java is a programing language”
OUTPUT: J I A P L

```
import java.util.Scanner;
public class Main {
```

```
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        String[] arr= str.split("\\s+");

        System.out.print("Initial letters in uppercase: ");
        for (int i = 0; i < arr.length; i++) {
            System.out.print(Character.toUpperCase(arr[i].charAt(0)) + " ");
        }
        obj.close();
    }
}
```

12. VOWELS AND CONSONANTS COUNT

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine().toLowerCase();
        int vowels = 0, consonants = 0;
        for (int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);
            if (ch >= 'a' && ch <= 'z') {
                if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                    vowels++;
                } else {
                    consonants++;
                }
            }
        }
        System.out.println("Vowels: " + vowels);
        System.out.println("Consonants: " + consonants);
    }
}
```

13. STRING COMPRESSION
INPUT : aaabbcddd
OUTPUT : a3b2c1d3

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
```

```
        Scanner obj = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str= obj.nextLine();
        obj.close();

        StringBuilder str2= new StringBuilder();
        for (int i = 0; i < str.length(); i++) {
            int count = 1;
            while (i + 1 < str.length() && str.charAt(i) == str.charAt(i + 1)) {
                i++;
                count++;
            }
            str2.append(str.charAt(i)).append(count);
        }
        System.out.println(str2);
    }
}
```

14. SWAP MAX AND MIN IN AN ARRAY

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args) {
        int[] arr = {3, 1, 4, 5, 2};
        int minIndex = 0, maxIndex = 0;
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] < arr[minIndex]) {
                minIndex = i;
            }
            if (arr[i] > arr[maxIndex]) {
                maxIndex = i;
            }
        }
        int temp = arr[minIndex];
        arr[minIndex] = arr[maxIndex];
        arr[maxIndex] = temp;
        for (int num : arr){
            System.out.print(num + " ");
        }
    }
}
```

15. GCD OF THE NUMBERS IN AN ARRAY

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n = obj.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = obj.nextInt();
        }
        int gcd = arr[0];
        for (int i = 1; i < n; i++) {
            int a = gcd, b = arr[i];
            while (b != 0) {
                int temp = b;
                b = a % b;
                a = temp;
            }
            gcd = a;
        }
        System.out.println("GCD of the array: " + gcd);
    }
}
```

16. CONVERT CAMEL CASE TO SNAKE CASE

Enter camelCase string: JavaIsSimple
Snake case: java_is_simple

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter camelCase string: ");
        String str= sc.nextLine();
        sc.close();
        String str2 = str.replaceAll("([a-z])([A-Z])", "$1_$2").toLowerCase();
        System.out.println("Snake case: " + str2);
    }
}
```

17. MATRIX MULTIPLICATION

```java
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows of Matrix A: ");
        int r1 = sc.nextInt();
        System.out.print("Enter number of columns of Matrix A (and rows of Matrix B): ");
        int c1 = sc.nextInt();

        System.out.print("Enter number of columns of Matrix B: ");
        int c2 = sc.nextInt();

        int[][] A = new int[r1][c1];
        int[][] B = new int[c1][c2];
        int[][] C = new int[r1][c2];
        System.out.println("Enter elements of Matrix A:");
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c1; j++) {
                A[i][j] = sc.nextInt();
            }
        }
        System.out.println("Enter elements of Matrix B:");
        for (int i = 0; i < c1; i++) {
            for (int j = 0; j < c2; j++) {
                B[i][j] = sc.nextInt();
            }
        }
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                C[i][j] = 0;
                for (int k = 0; k < c1; k++) {
                    C[i][j] += A[i][k] * B[k][j];
                }
            }
        }
        System.out.println("Resultant Matrix (AxB):");
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                System.out.print(C[i][j] + " ");
            }
            System.out.println();
        }
    }
```

```
}
```

18. Apple Redistribution into Boxes

```
import java.util.Scanner;
class Main{
   public static void main(String[] args){
      Scanner obj=new Scanner(System.in);
      System.out.println("Enter number of bundles: ");
      int bundle = obj.nextInt();
      System.out.println("Enter number of box: ");
      int box = obj.nextInt();


      int arr1[] = new int[bundle];
      int tot = 0;
      System.out.println("Enter apples in each bundles: ");
      for(int i=0;i<bundle;i++){
         arr1[i]= obj.nextInt();
         tot += arr1[i];
      }
      int arr2[] = new int[box];
      System.out.println("Enter accommodation of box: ");
      for(int i=0;i<box;i++){
         arr2[i]= obj.nextInt();
      }

      int max = arr2[0];
      int c=0,n=0;
      for(int j=0;j<box;j++){
         max = arr2[0];
      for(int i=0;i<box;i++){
         if(max<arr2[i]){
            max=arr2[i];
            n=i;
         }
      }
      arr2[n]=0;
      tot = tot-max;
      c++;
      if(tot<=0){
         break;
      }
      }
```

```
        System.out.println("Total box: " + c);
    }
}
```

19. STRING PALINDROME

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine().toLowerCase();
        String str2 = str.replaceAll("\\s", "");
        String str3 = new StringBuilder(str2).reverse().toString();

        if (str2.equals(str3)) {
            System.out.println("True");
        } else {
            System.out.println("False");
        }
    }
}
```

20. COUNT (A, AN, THE) IN A SENTENCE

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine().toLowerCase();
        String str2[] = str.split("\\s+");
        int an=0, a=0, the=0;
        for(String i : str2){
        if (i.equals("an")) {
            an++;
        }
        if (i.equals("a")) {
            a++;
        }
        if (i.equals("the")) {
            the++;
        }
        }
        System.out.println("An count: " + an);
```

```
        System.out.println("A count: " + a);
        System.out.println("The count: " + the);
    }
}
```

21. WORDS TO A SENTENCE

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        obj.nextLine();
        String str[] = new String[n];
        for(int i=0;i<n;i++){
            str[i]= obj.nextLine();
        }
        StringBuilder str2 = new StringBuilder() ;
        for(int i=0;i<n;i++){
            str2.append(str[i]);
            if(i<n-1){
                str2.append(" ");
            }
        }
        System.out.println(str2.toString());
    }
}
```

(or)

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        obj.nextLine();
        String str[] = new String[n];
        for(int i=0;i<n;i++){
            str[i]= obj.nextLine();
        }
        String str2 = String.join(" ", str);
        System.out.println(str2.toString());
    }
}
```

22. PRINT THE INDEX FOR THE PATTERN FOUND IN THE SENTENCE

Input : abcdefafgabcrf
abc
Output : 0,8

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first string: ");
        String text =sc.nextLine();
        System.out.print("Enter second string: ");
        String pattern = sc.nextLine();

        int index = text.indexOf(pattern); // Find first occurrence
        boolean found = false;

        while (index != -1) {
            System.out.print(index);
            found = true;
            index = text.indexOf(pattern, index + 1); // Find next occurrence

            if (index != -1) {
                System.out.print(",");
            }
        }
        if (!found) {
            System.out.println("-1");
        }
    }
}
```

23. ANAGRAM

INPUT: listen
Silent
OUTPUT: True

```
import java.util.Arrays;
import java.util.Scanner;
class Main {
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first string: ");
        String str1 = sc.nextLine();
        System.out.print("Enter second string: ");
        String str2 = sc.nextLine();
        str1 = str1.replaceAll("\\s", "").toLowerCase();
        str2 = str2.replaceAll("\\s", "").toLowerCase();

        if (str1.length() != str2.length()) {
            System.out.println("False");
        } else {
            char[] arr1 = str1.toCharArray();
            char[] arr2 = str2.toCharArray();
            Arrays.sort(arr1);
            Arrays.sort(arr2);

            if (Arrays.equals(arr1, arr2)) {
                System.out.println("True");
            } else {
                System.out.println("False");
            }
        }
    }
}
```

24. PANGRAM

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a sentence: ");
        String sentence = sc.nextLine();
        sentence = sentence.toLowerCase().replaceAll("[^a-z]", "");

        boolean[] alphabet = new boolean[26];
        int count = 0;
        for (char c : sentence.toCharArray()) {
            int index = c - 'a';
            if (!alphabet[index]) {
                alphabet[index] = true;
                count++;
            }
```

```
        }
        if (count == 26) {
            System.out.println("True");
        } else {
            System.out.println("False");
        }
    }
}
```

25. FIND THE FREQUENCY OF EACH WORD IN A SENTENCE

Input: "apple banana apple",
Output: apple - 2
banana - 1

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        String text = obj.nextLine();
        String[] words = text.split(" ");
        boolean[] counted = new boolean[words.length];

        for (int i = 0; i < words.length; i++) {
            if (counted[i]){
                continue;
            }
            int count = 1;
            for (int j = i + 1; j < words.length; j++) {
                if (words[i].equals(words[j])) {
                    count++;
                    counted[j] = true;
                }
            }
            System.out.println(words[i] + " - " + count);
        }
    }
}
```

26. PALINDROME COUNT IN AN ARRAY OF NUMBERS

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = scanner.nextInt();
        int[] numbers = new int[n];
        System.out.println("Enter the numbers:");
        for (int i = 0; i < n; i++) {
            numbers[i] = scanner.nextInt();
        }
        int count = 0;
        for (int num : numbers) {
            int original = num, reverse = 0;
            while (num > 0) {
                reverse = reverse * 10 + num % 10;
                num /= 10;
            }
            if (original == reverse) {
                count++;
            }
        }
        System.out.println("Number of palindromic numbers: " + count);
    }
}
```

27. PALINDROME - PS

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String str = obj.nextLine();

        String str2 = str.replaceAll("[^a-zA-Z]", "").toLowerCase();
        char[] chars = str2.toCharArray();
        int length = chars.length;
        boolean isPalindrome = true;

        for (int i = 0; i < length / 2; i++) {
            if (chars[i] != chars[length - i - 1]) {
                isPalindrome = false;
                break;
            }
        }
        if (isPalindrome) {
```

```
            System.out.println(str + " is a palindrome");
        } else {
            System.out.println(str + " is Not a palindrome");
        }
    }
}
```

28. CURRENT NUMBER TO THE POWER OF NEXT NUMBER - (last number will have the power of the first number)

INPUT: 5
1 2 3 4 5
OUTPUT: 1 8 81 1024 5
(1^2, 2^3, 3^4, 4^5,5^1)

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        int n = obj.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i]= obj.nextInt();
        }
        int[] result = new int[n];
        for (int i = 0; i < n - 1; i++) {
            result[i] = (int) Math.pow(arr[i], arr[i + 1]);
        }
        result[n - 1] = (int) Math.pow(arr[n - 1], arr[0]);

        for (int i : result) {
            System.out.print(i + " ");
        }
    }
}
```

29. REVERSE THE STRING

```
import java.util.Scanner;
public class Main
{
            public static void main(String[] args) {
                Scanner obj = new Scanner(System.in);
                String str = obj.nextLine();
```

```
            String[] arr = str.split("\\s+");
            int l = arr.length;
            for(int i=l-1;i>=0;i--){
                System.out.print(arr[i] + " ");
            }
        }
}
```

30. ADD THE EVEN NUMBERS AND MULTIPLY OTHERS

INPUT: 5

2 3 4 5 6

2 3 4 5 6

OUTPUT: 4 9 8 25 12

```
import java.util.Scanner;
public class Main
{
        public static void main(String[] args) {
            Scanner obj = new Scanner(System.in);
            int n = obj.nextInt();
            int a1[] = new int[n];
            int a2[] = new int[n];
            for(int i=0;i<n;i++){
                a1[i]=obj.nextInt();
            }
            for(int i=0;i<n;i++){
                a2[i]=obj.nextInt();
            }
            for(int i=0;i<n;i++){
                if(a1[i]%2 == 0 && a2[i]%2==0){
                    System.out.print(a1[i]+a2[i] +" ");
                }
                else{
                    System.out.print(a1[i]*a2[i]+ " ");
                }
            }
        }
}
```

31. CYCLIC STRING ROTATION

1. **INPUT: Hello world**

   3

   OUTPUT: lo worldHel

2. **INPUT: jpd**
   4
   OUTPUT: pdj

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner obj = new Scanner(System.in);
        String str = obj.nextLine();
        int n = obj.nextInt();
        n = n % str.length();
        String str2= str.substring(n) + str.substring(0, n);
        System.out.println(str2);
    }
}
```

32. LONGEST COMMON PREFIX

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.nextLine();
        String prefix = scanner.nextLine();
        for (int i = 1; i < n; i++) {
            String current = scanner.nextLine();
            while (!current.startsWith(prefix)) {
                prefix = prefix.substring(0, prefix.length() - 1);
                if (prefix.isEmpty()) {
                    System.out.println("");
                    return;
                }
            }
        }
        System.out.println(prefix);
    }
}
```

33. PRIME IN EVEN POSITION

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter how many prime numbers you want to generate: ");
        int numPrimes = scanner.nextInt();

        int count = 0;
        int num = 2;
        int primeCount = 0;

        while (primeCount < numPrimes) {
            if (isPrime(num)) {
                if (primeCount % 2 == 0) {
                    System.out.println(num);
                }
                primeCount++;
            }
            num++;
        }

        scanner.close();
    }

    public static boolean isPrime(int num) {
        if (num <= 1) return false;
        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0) {
                return false;
            }
        }
        return true;
    }
}
```